# Part 1: Exam day

## Exam day: what to know by heart

### The exam

- The exam is written and school-based: answer Question 1, then either Question 2 or Question 3. [Class 9, slide 33]
- The two answers you give count equally.
- No aids are allowed, and answers can be tested for plagiarism: exam papers, 4 December 2024 and 13 June 2025.
- The December 2024 paper allows four hours; the 2026 date is not published, so check the course plan. [Class 9, slide 33]

### Past exam questions

| Exam | Question | Themes |
|---|---|---|
| Dec 2024 Q1 | What do you understand by innovation system? What do you understand by sustainable transformation? What do you understand by "agency"? What do you understand by "institutions"? | TH1, TH3, TH7 |
| Dec 2024 Q2 | How can "agency" in innovation systems promote or inhibit sustainable transformation? Use examples. | TH5, TH8 |
| Dec 2024 Q3 | How can technologies in innovation systems promote or inhibit sustainable transformation? Use examples. | TH6, TH7 |
| Jun 2025 Q1 | What do you understand by a regional innovation system? What do you understand by institutional work? What do you understand by sustainability? | TH2, TH3, TH7 |
| Jun 2025 Q2 | In what way can a regional innovation system promote or inhibit industrial restructuring? Use examples. | TH2, TH5 |
| Jun 2025 Q3 | How can technological innovation systems be guided to promote sustainable transformation? Use examples. | TH6 |

### Compare the systems

| | National IS | Regional IS | Technological IS | Entrepreneurial ecosystem | Crowdfunding ecosystem |
|---|---|---|---|---|---|
| Focus | A country's learning and innovation | A region's knowledge and industry | One technology and its functions | New ventures and their context | Platforms and cocreated value |
| Boundary | The nation | The region | The technology | The ecosystem's actors | The platform ecosystem |
| Key actors | Firms, universities, the state | Regional firms, clusters, agencies | Firms, research, policy | Entrepreneurs, investors, mentors | Platforms, backers, creators |
| What drives change | Path dependency and policy | Related variety and institutional work | The seven functions | Selection and strategic choice | Value propositions |
| Norway example | Petroleum suppliers instead of Dutch disease | Hordaland and Rogaland | Longship and maritime batteries | Jahnebakken in Bergen | Norwegian crowdfunding growth |
| Key readings | P01, P02 | P03, P05 | P09, P10, P17 | P06, P07, P08 | P11, P12, P13 |

The comparison follows [Class 9, slide 25] and the cards.

### Must-know definitions

1. Innovation system: learning and innovation across the whole economy, not only universities. (Fagerberg et al. 2009, p. 431)
2. National innovation system: the country-level version, from Freeman, Lundvall and Nelson. (Fagerberg et al. 2009, p. 431)
3. Regional innovation system: citing Cooke, interacting subsystems of knowledge and commercial exploitation. (Fløysand et al. 2022, p. 16)
4. Technological innovation system: citing Carlsson, the agents around one technology. (Hekkert et al. 2007, p. 416)
5. Entrepreneurial ecosystem: citing Hawley, co-evolution, mutualistic interdependence and diverse, nested actors. (Stam & van de Ven 2021, p. 811)
6. Institutions: citing North, the formal and informal rules that shape action. (Fløysand et al. 2022, p. 16)
7. Institutional work: citing Lawrence, work that actors do on institutions. (Fløysand et al. 2022, p. 15)
8. Agency: following Bækkelund, change agency alters a path and reproductive agency keeps it going. (Bækkelund 2021, p. 759)
9. Path dependence: history that favours some solutions and holds others back. (Fagerberg et al. 2009, p. 431)
10. Path creation: citing MacKinnon, new development trajectories emerging in a region. (Njøs et al. 2020, p. 268)
11. Sustainable transformation: citing UNEP, change that cuts emissions and protects biodiversity. (Fløysand et al. 2022, p. 17)
12. Sustainability: ecological continuity, justice within and between generations, quality of life. (Schlaile et al. 2017, p. 5)

### How to answer

- Question 1: start with a direct definition (By X, I understand …) and show how the readings and classes use the term. [Class 9, slides 34–35]
- Question 2 or 3: introduction [Class 9, slide 39], main part [Class 9, slide 40], conclusion [Class 9, slide 41].
- What examiners look for [Class 9, slide 43]:
- [ ] Structure: introduction, theory, empirical material and conclusion, each with its weight.
- [ ] Problem statement: follow the question you were given, and show how you read it.
- [ ] Language and argument: independence, an argument carried through, a tidy presentation.
- [ ] Theory: the readings put to use, relevant and understood, fitted to the problem.
- [ ] Analysis: relevant cases, analysed critically and related to theory.
- [ ] Conclusion: back to the problem, the findings summed up, the limits stated.

### Promote or inhibit

- Promotes: related variety lets a region recombine what it has into a new path. (Fløysand & Jakobsen 2016, p. 330)
- Inhibits: a specialized, dirigiste system can keep a region in one industry. (Fløysand et al. 2022, p. 15)
- Promotes: a cluster and public procurement can create a market for a new technology. (Njøs et al. 2020, pp. 281–282)
- Inhibits: weak market formation and weak materialization hold a technology back. (Steen et al. 2024, p. 9)
- Promotes: collective institutional work can legitimize a green shift. (Fløysand et al. 2022, p. 15)
- Inhibits: institutional work can unintentionally sustain oil and gas. (Fløysand et al. 2022, p. 25)

# Part 2: Themes

## Theme 1: Innovation systems and Norway's national innovation system

### Core definitions

- **Innovation system** (*innovasjonssystem*): learning and innovation across the whole economy, not only in universities and research institutes. (Fagerberg et al. 2009, p. 431)
- **National innovation system** (*nasjonalt innovasjonssystem*): the same at the scale of a country, a term the authors trace to Freeman, Lundvall and Nelson. (Fagerberg et al. 2009, p. 431)
- **Institutions** (*institusjoner*): citing North, the rules of the game, costly to establish but scale-giving once adopted. (Fagerberg et al. 2009, p. 433)
- **Path dependency** (*stiavhengighet*): historical processes that favour some solutions and hold others back. (Fagerberg et al. 2009, p. 431)
- **Trinity of innovation** (*treenighet*): citing Schwarz and Thompson, the patterns of people, things and ideas behind technology. (Fløysand et al. 2013, PDF p. 9)

### Key models and figures

**Three industrial development paths** (Fagerberg et al. 2009, p. 439):

1. Small-scale decentralized path: small firms investing little of their own funds.
2. Large-scale centralized path: metals, electrochemistry and hydropower, often foreign-financed.
3. Knowledge-intensive network-based path: R&D-intensive firms in sectors such as ICT.

**The pragmatic innovation complex** (Fløysand et al. 2013, PDF p. 10):

1. Company and tools meet through technique.
2. Tools and tasks meet through production, with unintended outputs.
3. Company and tasks meet through rationale, profit.

### Exam angles

- Dec 2024 Q1: the four terms, each defined and tied to Norway, with one source per term. (our reading)
- Class 9 example: how innovation systems promote or inhibit sustainable transformation. [Class 9, slide 38]
- Practice: use the petroleum suppliers as the example. (ours)

### Norway examples

- **Domestic petroleum suppliers**: shipbuilding, engineering and services grew with petroleum instead of Dutch disease. (Fagerberg et al. 2009, p. 434)
- **The concession laws**: early waterfall rules became the model for national control of offshore oil and gas, decades later. (Fagerberg et al. 2009, p. 441)
- **Fish-farming**: disease control was mastered without a large corporate laboratory, building on regional resources. (Fagerberg et al. 2009, p. 441)

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** The question asks for four terms, and I answer by defining each one and tying it to the Norwegian case.
- **Theory:** Fagerberg et al. (2009) use a broad definition of the national innovation system, where learning happens across the whole economy. Citing North, they treat institutions as the rules of the game, and they show how path dependency keeps a system on its track. Fløysand et al. (2013) add the trinity of actors, objects and ideas.
- **Example:** Fagerberg et al. (2009) show that petroleum did not bring Dutch disease, because public policy helped domestic suppliers in shipbuilding and engineering grow with the industry. The same history shows path dependency, since the concession laws of the waterfalls became the model for controlling offshore oil.
- **Mini-conclusion:** An innovation system is therefore a country's whole way of learning, ruled by institutions and carried by its history.

### Links

Cards: P01, P02. Themes: TH2, TH3.

## Theme 2: Regional innovation systems and industrial restructuring

### Core definitions

- **Regional innovation system** (*regionalt innovasjonssystem*): citing Cooke, interacting subsystems of knowledge infrastructure and commercial exploitation, set in knowledge, industry and governance. (Fløysand et al. 2022, p. 16)
- **Related variety** (*relatert variasjon*): industries that differ but are related, which gives a region room to combine them and innovate across industries. (Fløysand & Jakobsen 2016, p. 330)
- **Path extension** (*baneforlengelse*): more of the same, continuing an existing industrial path rather than opening a new one. (Fløysand & Jakobsen 2016, p. 328)
- **Path creation** (*baneskaping*): developing a new industry rather than extending an old one. (Fløysand & Jakobsen 2016, p. 328)
- **Institutional work** (*institusjonelt arbeid*): citing Lawrence and colleagues, the work actors do to affect institutions, whether maintaining, creating, changing or disrupting them. (Fløysand et al. 2022, p. 15)

### Key models and figures

**Three levels of regional renewal** (Fløysand & Jakobsen 2016, p. 324):

1. Industry level: cluster dynamics, how firms and support institutions interact.
2. Firm level: related variety across industries.
3. Individual level: entrepreneurship, the actors who start something new.

**Four industrial trajectories** (Fløysand & Jakobsen 2016, p. 328):

1. Path exhaustion: an industry weakens and loses its base.
2. Path extension: more of the same.
3. Path renewal: upgrading the industry with new products.
4. Path creation: a new industry.

### Exam angles

- Jun 2025 Q2: how a regional system promotes or inhibits restructuring, with examples from the two regions. (our reading)
- Jun 2025 Q1: the regional system and institutional work, defined through the two regions. (our reading)
- Practice: compare Hordaland and Rogaland on the same question, term by term. (ours)

### Norway examples

- **Hordaland**: a diversified, interactive system where collective work legitimizes a green shift. (Fløysand et al. 2022, p. 15)
- **Rogaland**: a specialized, dirigiste system where individual work delegitimizes it. (Fløysand et al. 2022, p. 15)
- **Mongstad microalgae**: a pilot that would join oil knowledge to marine knowledge in fish feed. (Fløysand & Jakobsen 2016, p. 337)

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how a regional innovation system can help or hinder industrial restructuring, and I answer with the concept, related variety and two Norwegian regions, Hordaland and Rogaland.
- **Theory:** Cooke's definition treats a regional innovation system as interacting subsystems of knowledge and exploitation. Fløysand and Jakobsen (2016) add related variety: industries that differ but are related give a region room to combine them, and so to renew without starting from nothing.
- **Example:** Fløysand et al. (2022) show that Hordaland's diversified clusters legitimized a green shift, while Rogaland's specialized petroleum system kept its path. The same comparison shows that a thick, specialized system can hold a region to one path.
- **Mini-conclusion:** A regional system promotes restructuring when related variety and collective work open a new path, and inhibits it when one industry holds the region in place.

### Links

Cards: P03, P05. Themes: TH1, TH3, TH5.

## Theme 3: Institutions and institutional work

### Core definitions

- **Institutions** (*institusjoner*): citing North, the formal and informal rules that shape action, both written rules and informal ways of doing. (Fløysand et al. 2022, p. 16)
- **Institutional work** (*institusjonelt arbeid*): citing Lawrence and colleagues, the physical and mental work actors do to affect institutions, whether maintaining, creating, changing or disrupting them. (Fløysand et al. 2022, p. 15)
- **Institutional logics** (*institusjonelle logikker*): citing Thornton and Almandoz, practice patterns and social prescriptions that legitimize practice, and so direct what actors see as possible. (Fløysand et al. 2022, p. 18)
- **Change agency** (*endringsaktørskap*): following Bækkelund, the agency that alters a path. (Fløysand et al. 2022, p. 17)
- **Reproductive agency** (*reproduktivt aktørskap*): following Bækkelund, the agency that keeps a path going. (Fløysand et al. 2022, p. 17)

### Key models and figures

**Four forms of institutional work**, citing Lawrence and colleagues (Fløysand et al. 2022, p. 17):

1. Maintaining institutions: keeping rules in place.
2. Creating new institutions: building rules that did not exist.
3. Changing existing institutions: reshaping what is there.
4. Disrupting institutions: breaking them down.

**The lock-in cycle in Rogaland** (Fløysand et al. 2022, p. 24):

1. A recession lowers costs in the petroleum sector.
2. Oil prices recover.
3. High profit margins return to the sector.
4. The sector bounces back and the region stays on its path.

### Exam angles

- Jun 2025 Q1: institutional work, defined and shown in the two regions. (our reading)
- Dec 2024 Q1: institutions as one of the four terms to define, with North. (our reading)
- Practice: use Arne Rettedal as individual leadership, and the lock-in cycle. (ours)

### Norway examples

- **Hordaland**: collective leadership and cross-cluster work legitimize a green shift and support clustering. (Fløysand et al. 2022, p. 15)
- **Rogaland**: individual leadership, linked to Arne Rettedal and the petroleum headquarters in Stavanger. (Fløysand et al. 2022, p. 22)
- **The dominance that lasts**: oil and gas still dominates both regions, unintentionally, despite a shared green vision. [Class 9, slide 31]

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** The question asks what institutional work means, and I answer with the definition, the two regions, one example and a short conclusion.
- **Theory:** Citing Lawrence and colleagues, Fløysand et al. (2022) define institutional work as the physical and mental work that actors do to affect institutions. Those efforts can maintain, create, change or disrupt them, and they work through institutional logics that decide what counts as legitimate.
- **Example:** Fløysand et al. (2022) show that collective work in Hordaland legitimized a green shift, while in Rogaland individual leadership kept the petroleum path. In both regions, oil and gas still dominates, which the authors call an unintended outcome.
- **Mini-conclusion:** Institutional work is therefore the everyday effort that keeps a regional system in place or moves it, often without anyone intending the result.

### Links

Cards: P03, P01. Themes: TH2, TH5, TH7.

## Theme 4: Entrepreneurial ecosystems

### Core definitions

- **Entrepreneurial ecosystem** (*entreprenørielt økosystem*): citing Hawley, a system marked by co-evolution, mutualistic interdependence and diverse, nested actors, which begins when entrepreneurs' paths cross. (Stam & van de Ven 2021, p. 811)
- **Munificence** (*rikelighet*): how many resources an ecosystem holds, from finance to talent. (Spigel & Harrison 2018, p. 161)
- **Functionality** (*funksjonalitet*): how easily entrepreneurs can reach those resources, through networks and trust. (Spigel & Harrison 2018, p. 161)
- **Selection effects** (*seleksjonseffekter*): opportunity costs and social legitimacy decide who becomes an entrepreneur, before any venture starts, through opportunity costs. (Autio et al. 2014, p. 1100)
- **Strategic choice effects** (*strategiske valgeffekter*): what those who enter then judge desirable and feasible, once they have started. (Autio et al. 2014, p. 1100)

### Key models and figures

**Ten elements in three groups** (Stam & van de Ven 2021, p. 813):

1. Institutional arrangements: formal institutions, culture and networks that set the rules.
2. Resources: infrastructure, finance, leadership, talent, knowledge, services and demand.
3. Output: productive entrepreneurship, the firms that commercialise and grow.

**Four ecosystem types** (Spigel & Harrison 2018, p. 163):

1. Strong: many resources, well-functioning networks, as in Silicon Valley.
2. Arid: sparse resources, well-functioning networks, as in some developing economies.
3. Irrigated: many resources, poorly functioning networks.
4. Weak: sparse resources, poorly functioning networks, as in deindustrialised regions.

### Exam angles

- Practice: how an ecosystem promotes or inhibits sustainable transformation, through the ten elements. (ours)
- Dec 2024 Q2: agency read through selection and strategic choice, with a concrete example. (our reading)
- Jun 2025 Q1: an ecosystem defined beside an innovation system. (our reading)

### Norway examples

- **Jahnebakken in Bergen**: an investor and founder house that gathers capital, mentors and new firms in one house. [Class 4, slide 19]
- **Work visas**: Norway uses work visas as one immigration-policy lever for entrepreneurs, beside start-up visas in other countries. [Class 4, slide 17]

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how an ecosystem can help or hinder sustainable transformation, and I answer with the elements, two effects and a Norwegian example from Bergen.
- **Theory:** Stam and van de Ven (2021) sort the ecosystem into institutions, resources and output. Autio et al. (2014) add selection and strategic choice, which decide who enters and what they attempt, and so shape the rate of new ventures.
- **Example:** In Bergen, an investor and founder house gathers capital, mentors and new firms in one place, which raises the resources an entrepreneur can reach and shortens the distance to them [Class 4, slide 19]. A house like that can also narrow the field, if only a few investors decide what gets funded each year.
- **Mini-conclusion:** An ecosystem promotes transformation when its resources are rich and easy to reach, and inhibits it when access depends on a few gatekeepers who decide what gets funded each year.

### Links

Cards: P06, P07, P08. Themes: TH1, TH5.

## Theme 5: Agency and new path development

### Core definitions

- **Agency** (*aktørskap*): citing Sotarauta and Suvinen, action or intervention by an actor to produce a particular effect, whether intended or not. (Bækkelund 2021, p. 758)
- **Change agency** (*endringsaktørskap*): following Bækkelund, the agency that alters a path, through entrepreneurship, institutional work or leadership. (Bækkelund 2021, p. 759)
- **Reproductive agency** (*reproduktivt aktørskap*): following Bækkelund, the agency that keeps a path going, through replication, maintenance or institutional work. (Bækkelund 2021, p. 759)
- **Path creation** (*baneskaping*): citing MacKinnon and colleagues, the emergence of new development trajectories in a region, such as a new industry. (Njøs et al. 2020, p. 268)
- **Bricolage** (*bricolage*): citing Levi-Strauss, resourcefulness and improvisation with what is at hand, in modest and steady steps. (Garud & Karnøe 2003, p. 278)

### Key models and figures

**A range from change to reproductive agency** (Bækkelund 2021, p. 760):

1. Change: Schumpeterian entrepreneurship, new combinations.
2. Change: institutional entrepreneurship, rewriting the rules.
3. Change: place-based leadership, rooted in the place.
4. Reproductive: replicative entrepreneurship, copying what works.
5. Reproductive: institutional work.
6. Reproductive: maintenance leadership, keeping the place steady.

**Three modes of change** (Miörner & Trippl 2017, p. 486):

1. Layering: adding something new to the support structure.
2. Adaptation: changing what exists.
3. Novel application: reusing what exists in a new way.

### Exam angles

- Dec 2024 Q2: agency promotes or inhibits through change and reproductive work. (our reading)
- Class 9 example: how innovation systems promote or inhibit sustainable transformation. [Class 9, slide 38]
- Practice: compare Flåm and Sogndal on the same range of agency and modes of change. (ours)

### Norway examples

- **Flåm, Odda and Sogndal**: three fjord villages with the same institutions and landscape but different tourism paths, a most-similar design. (Bækkelund 2021, p. 761)
- **Flåm**: more change agency than the other two villages. (Bækkelund 2021, p. 765)
- **Maritime batteries**: a cluster set a direction and public procurement created the market. (Njøs et al. 2020, pp. 281–282)

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how agency can push a system towards sustainability or hold it back, and I answer with two kinds of agency and one Norwegian case, the three fjord villages.
- **Theory:** Bækkelund (2021) separates change agency, which alters a path, from reproductive agency, which keeps it going. Garud and Karnøe (2003) show that agency is distributed, so no single hero carries a new path alone.
- **Example:** Bækkelund (2021) shows that in the three fjord villages reproductive work was stronger before a path formed and change work stronger in path creation. Flåm shows more change agency than Odda and Sogndal, which shows that the same landscape does not decide the outcome.
- **Mini-conclusion:** Agency promotes transformation when change work finds a critical juncture, and inhibits it when reproductive work keeps the old path comfortable and unquestioned.

### Links

Cards: P14, P15, P16, P17. Themes: TH2, TH3, TH6.

## Theme 6: Technological innovation systems and their functions

### Core definitions

- **Technological innovation system** (*teknologisk innovasjonssystem*): citing Carlsson and Stankiewicz, a network of agents that create, spread and use a technology under shared institutions. (Hekkert et al. 2007, p. 416)
- **Function** (*funksjon*): an activity that results in technological change; Hekkert and colleagues list seven. (Hekkert et al. 2007, p. 415)
- **Guidance of the search** (*retning for søket*): the function that sets expectations and goals and so steers where a technology goes. (Hekkert et al. 2007, p. 423)
- **Materialization** (*materialisering*): the function that covers large physical infrastructure, added because CCS needs it. (Steen et al. 2024, p. 2)
- **Legitimation** (*legitimering*): the function that covers acceptance and legitimacy. (Steen et al. 2024, p. 3)

### Key models and figures

**Seven functions of innovation systems** (Hekkert et al. 2007, pp. 421–425):

1. Entrepreneurial activity
2. Knowledge development
3. Knowledge diffusion through networks
4. Guidance of the search
5. Market formation
6. Resource mobilization
7. Creation of legitimacy

**Three motors of change** (Hekkert et al. 2007, p. 426):

1. Motor A: lobbying for market formation, then entrepreneurial activity.
2. Motor B: lobbying for research resources, then higher expectations.
3. Motor C: guidance of the search, then resources and knowledge.

### Exam angles

- Jun 2025 Q3: rate each function as strong, intermediate or weak, then aim policy at the weak ones. (our reading)
- Dec 2024 Q3: technologies promote or inhibit through the functions that are strong or weak. (our reading)
- Class 9 example: a TIS and the seven functions. [Class 9, slide 37]
- Practice: compare CCS and maritime batteries with the same seven functions and see which differ. (ours)

### Norway examples

- **Longship and Northern Lights**: the main exceptions to weak materialization: large projects now being built with state support. (Steen et al. 2024, p. 10)
- **Maritime batteries**: a cluster set a direction, public procurement created the market, and some seventy ferry lines followed. (Njøs et al. 2020, pp. 281–282)
- **Mongstad**: an earlier plan for a full-scale plant at the refinery and power station. (Steen et al. 2024, p. 4)

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how actors can steer a technological innovation system towards sustainability, and I answer with the functions, a Norwegian case and a conclusion.
- **Theory:** Hekkert et al. (2007) list seven functions, and guidance of the search is the one that sets direction. Steen et al. (2024) show how the functions can be rated as strong, intermediate or weak across a whole system.
- **Example:** Steen et al. (2024) describe Longship and Northern Lights as the main exceptions to weak materialization. The same case shows weak market formation, because costs are high and the emissions trading system does not yet justify investment.
- **Mini-conclusion:** A system can be guided when policy targets the functions that stay weak.

### Links

Cards: P09, P10, P17. Themes: TH5, TH7.

## Theme 7: Sustainability, transformation and normative direction

### Core definitions

- **Green industrial restructuring** (*grønn industriell omstilling*): citing UNEP, change that cuts emissions and pollution, raises resource efficiency and protects biodiversity, such as a shift from petroleum. (Fløysand et al. 2022, p. 17)
- **Paradigm** (*paradigme*): a set of assumptions, concepts, values and practices that make up a shared worldview and span a bounded space of what counts as reasonable. (Schlaile et al. 2017, p. 4)
- **Dedicated innovation system** (*dedikert innovasjonssystem*): from Pyka, an innovation system dedicated to transformations towards sustainability, which puts trajectories under a sustainability goal. (Schlaile et al. 2017, p. 4)
- **Transformative innovation** (*transformativ innovasjon*): citing Steward, novelty that breaks the paradigm and changes the whole system, beyond new products or processes. (Schlaile et al. 2017, p. 5)
- **Transformational failures** (*transformasjonssvikt*): citing Weber and Rohracher, the failures that a sustainability policy must also address, such as weak direction, legitimacy or responsibility. (Schlaile et al. 2017, p. 5)

### Key models and figures

**Three normative questions** (Schlaile et al. 2017, pp. 5–7):

1. Directionality: what to transform or sustain.
2. Legitimacy: why, and who decides.
3. Responsibility: by and for whom.

### Exam angles

- Dec 2024 Q3: a technology promotes or inhibits only inside a direction, with one example of each side. (our reading)
- Jun 2025 Q1: sustainability as a normative term, not a neutral one, beside innovation system and regional system. (our reading)
- Practice: use the two petroleum regions as a case of a direction without agreement, and name who decides. (ours)

### Norway examples

- **The end of cheap growth**: rapid growth from offshore oil and gas is ending, and so is cheap hydropower, which the authors treat as a warning. (Fagerberg et al. 2009, p. 442)
- **The dominance that lasts**: oil and gas still dominates Hordaland and Rogaland, unintentionally, despite a shared green vision, which the authors call an unintended outcome. [Class 9, slide 31]

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how a technology can promote or hold back sustainable transformation, and I answer with the paradigm, the three questions and a Norwegian case from the two petroleum regions.
- **Theory:** Schlaile et al. (2017) argue that a shift at the paradigm level is the only way to integrate the normativity of sustainability. A dedicated system then asks what to transform, who decides and who answers for the result, in that order.
- **Example:** Fløysand et al. (2022) show that Hordaland and Rogaland both aim at a green shift, yet oil and gas still dominates their business. The same regions show that a technology promotes transformation only when the institutional work around it points the same way.
- **Mini-conclusion:** A technology promotes transformation when it sits inside a dedicated system, and inhibits it when it only extends the old paradigm and its interests.

### Links

Cards: P04, P03. Themes: TH1, TH3, TH6.

## Theme 8: Crowdfunding and service ecosystems

### Core definitions

- **Crowdfunding** (*folkefinansiering*): collective financing that also works as an instrument to foster innovation, and lets users act like venture capitalists. (Bargoni et al. 2024, p. 1110)
- **Incremental innovativeness** (*inkrementell innovativitet*): citing Chan and Parhankangas, a cumulative improvement of existing technology and capabilities, which builds on what a firm already does. (Bargoni et al. 2024, p. 1115)
- **Radical innovativeness** (*radikal innovativitet*): citing Chan and Parhankangas, change that is more drastic, unique and comprehensive, and opens something genuinely new. (Bargoni et al. 2024, p. 1115)
- **Value proposition** (*verdiforslag*): citing Frow and colleagues, a mechanism that adjusts as actors negotiate how resources are shared, so the offer changes as the relationship changes. (Quero & Ventura 2019, p. 52)
- **Crowdfunding ecosystem** (*folkefinansieringsøkosystem*): a service ecosystem in which actors cocreate value to carry one actor's proposal forward, which makes backers, platforms and institutions one system. (Quero & Ventura 2019, p. 48)

### Key models and figures

**Five steps of value cocreation**, adapting Frow and Payne (Quero & Ventura 2019, p. 51):

1. Identify actors.
2. Determine core values.
3. Facilitate dialogue between the actors.
4. Identify cocreation opportunities.
5. Cocreate value propositions with the actors.

**Six dimensions of each platform** (Dematos Guimarães et al. 2024, p. 8):

1. Country.
2. Geographical scope.
3. Business model.
4. Focus of the platform.
5. Institutional structure.
6. Foundation year.

### Exam angles

- Dec 2024 Q2: crowdfunding communities as one form of agency, marked as our reading. (our reading)
- Jun 2025 Q1: whether a crowdfunding ecosystem is also an innovation system, term by term. (our reading)
- Practice: read one campaign as a service ecosystem, actor by actor, and name who decides. (ours)

### Norway examples

- **Norwegian crowdfunding**: it grew from 0.77 to 892 million kroner between 2012 and 2020, more than a thousand times. [Bonus, slide 42]
- Only one Norwegian example is verified in the course material.

### Answer template

Drafting aid (ours): learn the moves, then write your own words.

- **Intro:** I read the question as asking how agency can promote or inhibit sustainable transformation, and I answer with crowdfunding communities, one ecosystem, one platform case and a conclusion.
- **Theory:** Bargoni et al. (2024) treat crowdfunding as more than money: it finances innovation at lower cost and tests projects outside the organisation. Quero and Ventura (2019) add that its actors cocreate value propositions, so backers are not only customers but partners in the offer.
- **Example:** Quero and Ventura (2019) show that two Spanish platform directors added public institutions to the actors of the ecosystem. The same case shows that crowdfunding can be read as agency, because backers decide together which projects go ahead, and that decision is the agency.
- **Mini-conclusion:** Crowdfunding communities promote transformation when they move resources to new projects, and inhibit it when only a few platforms set the terms and pick the winners before any campaign starts.

### Links

Cards: P11, P12, P13. Themes: TH1, TH5.

# Part 3: Paper cards

### Paper 1: Fagerberg, Mowery & Verspagen (2009)

*The evolution of Norway's national innovation system*, Science and Public Policy 36(6).

**In one line**

Norway grew fast with little research; the authors explain why through system, history and structure.

**Key concepts**

- **National innovation system** (*nasjonalt innovasjonssystem*): all learning and innovation across an economy, not only universities and laboratories; the authors trace it to Freeman, Lundvall and Nelson. (p. 431)
- **Institutions** (*institusjoner*): citing North, the "rules of the game", costly to establish but scale-giving once adopted. (p. 433)
- **Path dependency** (*stiavhengighet*): historical processes that favour some solutions and hold others back, say the authors citing Arthur. (p. 431)
- **Absorptive capacity** (*absorpsjonskapasitet*): citing Cohen and Levinthal, the ability to identify, absorb and use new knowledge. (p. 437)
- **Dutch disease** (*hollandsk syke*): currency appreciation after a resource discovery, which can bring de-industrialization. (p. 434)

**Model**

**Three industrial development paths** (p. 439):

1. Small-scale decentralized path: small firms investing little of their own funds.
2. Large-scale centralized path: metals, electrochemistry and hydropower, often foreign-financed.
3. Knowledge-intensive network-based path: R&D-intensive firms in sectors such as ICT.

**Findings**

- Adjusted for industrial structure, Norwegian firms do as much business research as peers abroad in five of six comparisons. (p. 436)
- The authors name knowledge diffusion and cooperation in innovation as the system's strong point, which standard indicators miss. (p. 438)
- The modernizers' computer research failed commercially, but its competences paid off in oil and gas. (p. 440)

**Norway link**

- Domestic suppliers in shipbuilding, engineering and services grew with petroleum, instead of Dutch disease. (p. 434)

**Exam use**

- Dec 2024 Q1: take the broad definition and institutions as rules of the game. (our reading)

**Limits**

- The authors leave aside the social and cultural roots of the system's adaptability. (p. 442)

**Hook**

Broad system, slow history, low research, high income.

### Paper 2: Fløysand, Jakobsen & Hidle (2013)

*The "trinity" of innovation: actors, ideas and objects in innovation practice*, Conference paper, 5th Nordic Geographers' Meeting, Reykjavík (paper in preparation, do not quote).

**In one line**

The paper asks how actors, ideas and objects meet in practice, and answers with a typology.

**Key concepts**

- **Trinity of innovation** (*treenighet*): citing Schwarz and Thompson, patterns of people, things and ideas behind technology. (PDF p. 9)
- **Actant** (*aktant*): citing Latour, any entity that acts or is treated as acting, nature included. (PDF p. 9)
- **Necessary relations** (*nødvendige relasjoner*): citing Sayer, the abstract properties of innovation that hold in any setting. (PDF p. 3)
- **Contingent relations** (*kontingente relasjoner*): cultural economy, history, governance and territory, which give the necessary relations content. (PDF p. 12)
- **Pragmatic innovation complex** (*pragmatisk innovasjonskompleks*): firms as actors, tools as the objects, and tasks as the ideas. (PDF p. 10)

**Model**

**The pragmatic innovation complex** (PDF p. 10, Figure 1):

1. Company and tools meet through technique.
2. Tools and tasks meet through production, with unintended outputs.
3. Company and tasks meet through rationale, profit.

**Findings**

- Innovation is discursive, material and practical, and the material side must not be lost. (PDF p. 14)
- Profit and productivity govern innovation in capitalist economies, but other interests count. (PDF p. 12)
- Public authorities and environmental groups can push innovation towards emission cuts. (PDF p. 13)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Dec 2024 Q1: the trinity and actants define innovation practice. (our reading)
- Jun 2025 Q1: necessary and contingent relations separate general from regional. (our reading)

**Limits**

- The authors do not judge whether innovations work well, only how. (PDF p. 13)
- Necessary relations alone cannot capture interaction, so qualitative study is needed. (PDF p. 13)

**Hook**

Teams, tools and tasks in one triangle.

### Paper 3: Fløysand et al. (2022)

*Institutional work, regional key actors, and green industrial restructuring*, Norsk Geografisk Tidsskrift – Norwegian Journal of Geography 76(1).

**In one line**

Two oil regions share a green vision, but institutional work legitimizes it in Hordaland and delegitimizes it in Rogaland.

**Key concepts**

- **Institutional work** (*institusjonelt arbeid*): citing Lawrence and colleagues, the physical and mental work actors do to affect institutions. (p. 15)
- **Regional innovation system** (*regionalt innovasjonssystem*): citing Cooke, interacting subsystems of knowledge generation and exploitation. (p. 16)
- **Institutions** (*institusjoner*): citing North, formal and informal rules that shape action. (p. 16)
- **Institutional logics** (*institusjonelle logikker*): citing Thornton and Almandoz, practice patterns and social prescriptions that legitimize practice. (p. 18)
- **Change and reproductive agency** (*endringsaktørskap og reproduktivt aktørskap*): following Bækkelund, work that alters or keeps a path. (p. 17)

**Model**

**Four forms of institutional work**, citing Lawrence and colleagues (p. 17):

1. Maintaining institutions: keeping rules and practices in place.
2. Creating institutions: building new rules and organisations.
3. Changing institutions: reshaping existing rules.
4. Disrupting institutions: breaking them down.

**Findings**

- In diversified Hordaland, collective-oriented work legitimizes a green shift. (p. 15)
- In petroleum-specialized Rogaland, individual-oriented work delegitimizes it. (p. 15)
- Institutional work can also sustain oil and gas, against intention. (p. 25)

**Norway link**

- **Hordaland and Rogaland**: two adjacent petroleum regions with opposite outcomes. (p. 15)
- **Hordaland crossover**: an intermediary says forty to fifty oil firms now also supply ocean farming. (p. 23)

**Exam use**

- Jun 2025 Q1: institutional work and a regional system, through the two cases. (our reading)
- Jun 2025 Q2: Hordaland promotes restructuring, Rogaland inhibits it. (our reading)

**Limits**

- The regions differ in history; maritime and salmon farming are stronger in Hordaland. (p. 25)
- More dialogue with interviewees might have raised validity. (p. 19)

**Hook**

Same vision, two regions, opposite outcomes.

### Paper 4: Schlaile et al. (2017)

*Innovation systems for transformations towards sustainability? Taking the normative dimension seriously*, Sustainability 9(12).

**In one line**

Innovation systems need a paradigm shift, the authors argue, before they can serve sustainability.

**Key concepts**

- **Innovation system** (*innovasjonssystem*): citing Niosi and colleagues, multiple interacting agents whose work produces, develops, protects, finances or regulates science and technology. (p. 2)
- **Paradigm** (*paradigme*): a set of assumptions, concepts, values and practices that make up a shared worldview. (p. 4)
- **Dedicated innovation system** (*dedikert innovasjonssystem*): from Pyka, a system dedicated to transformations towards sustainability. (p. 4)

- **Transformative innovation** (*transformativ innovasjon*): citing Steward, novelty that breaks the paradigm and changes the whole system. (p. 5)
- **Transformational failures** (*transformasjonssvikt*): citing Weber and Rohracher, the failures that a sustainability policy must also address. (p. 5)

**Model**

**Three normative questions** (pp. 5–7):

1. Directionality: what to transform or sustain.
2. Legitimacy: why, and who decides.
3. Responsibility: by and for whom.

**Findings**

- Technological fixes cannot solve wicked problems; economy, institutions, technology, culture and organisation must all change. (p. 2)
- The existing strands are scattered; only a shift at paradigm level can integrate sustainability's normativity. (p. 10)
- A dedicated system can in principle put innovation trajectories under a sustainability goal. (p. 10)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Dec 2024 Q1: sustainable transformation as a normative, directed shift. (our reading)
- Dec 2024 Q3: technologies alone cannot set a direction. (our reading)

**Limits**

- The authors close with Sendzimir's warning that no analysis will ever remove surprise and uncertainty. (p. 10)

**Hook**

A system that knows what it is for.

### Paper 5: Fløysand & Jakobsen (2016)

*Business development in Western Norway: adaptability and regional renewal (in Norwegian)*, Book chapter in Region og regionalisering, Novus Forlag.

**In one line**

Western Norway renews through related variety and entrepreneurship, while mature clusters mostly extend what they do.

**Key concepts**

- **Regional innovation system** (*regionalt innovasjonssystem*): citing Cooke and colleagues, interacting subsystems of knowledge infrastructure and commercial exploitation. (p. 326)
- **Related variety** (*relatert variasjon*): industries that differ but are related, which gives a region room to combine them. (p. 330)
- **Path extension** (*baneforlengelse*): more of the same, continuing an existing industrial path. (p. 328)
- **Path creation** (*baneskaping*): developing a new industry. (p. 328)

**Model**

**Three levels of regional renewal** (p. 324):

1. Industry level: cluster dynamics.
2. Firm level: related variety.
3. Individual level: entrepreneurship.

**Findings**

- Specialisation can breed continuity and stagnation, so related industries and international networks matter. (p. 324)
- Ten of eighteen western economic areas score above the national median for related variety. (p. 337)
- The region's share of national secondary employment rose from 24 to 33 percent between 1970 and 2010. (pp. 331–332)

**Norway link**

- **Mongstad microalgae**: a pilot that would use carbon dioxide from oil and gas to grow fish feed. (p. 337)
- **The cluster programme**: Western Norway's clusters are recognised at Arena, NCE and GCE levels. (p. 333)

**Exam use**

- Jun 2025 Q2: related variety can promote restructuring, while specialisation can inhibit it. (our reading)

**Limits**

- The Ulstein interview came from a cluster study and was not designed for this question. (p. 340)
- Which driver matters most cannot be settled. (p. 343)

**Hook**

Four counties, three levels, one path bending.

### Paper 6: Spigel & Harrison (2018)

*Toward a process theory of entrepreneurial ecosystems*, Strategic Entrepreneurship Journal 12(1).

**In one line**

The authors turn the entrepreneurial ecosystem from a policy slogan into a process through which entrepreneurs acquire resources.

**Key concepts**

- **Entrepreneurial ecosystem** (*entreprenørielt økosystem*): better understood as an ongoing process, not a thing, through which entrepreneurs acquire resources, knowledge and support. (p. 158)
- **Chaotic conception** (*kaotisk begrep*): citing Sayer, a concept that describes rather than explains, which the ecosystem idea had become, and the authors want to give it a theory. (p. 152)
- **Munificence** (*rikelighet*): how many resources an ecosystem holds. (p. 161)
- **Functionality** (*funksjonalitet*): how easily entrepreneurs can reach those resources. (p. 161)
- **Recycling** (*resirkulering*): founders who return as investors and mentors, and workers who carry skills to other ventures. (p. 160)

**Model**

**Four ecosystem types** (p. 163, Figure 2):

1. Strong: many resources, well-functioning networks.
2. Arid: sparse resources, well-functioning networks.
3. Irrigated: many resources, poorly functioning networks.
4. Weak: sparse resources, poorly functioning networks.

**Findings**

- Access to resources matters more than their presence. (p. 159)
- Recycling depends on a culture that does not punish failure. (p. 160)
- Strong, well-functioning ecosystems should see more growth-oriented entrepreneurship. (p. 163)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Practice: an ecosystem can promote transformation by giving access, or inhibit it by also holding those resources back. (our reading)

**Limits**

- The propositions are offered to guide research, not tested here. (p. 165)
- Defining ecosystems by their activity levels would be circular. (p. 164)

**Hook**

A process, not a place.

### Paper 7: Stam & van de Ven (2021)

*Entrepreneurial ecosystem elements*, Small Business Economics 56.

**In one line**

The authors define the ecosystem, give it ten elements, and turn them into an index for the twelve Dutch provinces.

**Key concepts**

- **Entrepreneurial ecosystem** (*entreprenørielt økosystem*): citing Hawley, a system marked by co-evolution, mutualistic interdependence and diverse, nested actors; it begins when entrepreneurs' paths cross. (p. 811)
- **Productive entrepreneurship** (*produktivt entreprenørskap*): citing Baumol, the firms that commercialise innovations and grow, the model's output. (p. 813)
- **Partisan mutual adjustment** (*partisk gjensidig tilpasning*): citing Lindblom, coordination without a coordinator. (p. 812)
- **Upward and downward causation** (*oppadgående og nedadgående kausalitet*): the elements explain activity, and activity feeds back into the elements. (p. 814)
- **Tautology** (*tautologi*): inferring a good ecosystem from the success it is meant to explain. (p. 810)

**Model**

**Ten elements in three groups** (p. 813):

1. Institutional arrangements: formal institutions, culture and networks.
2. Resources: infrastructure, finance, leadership, talent, knowledge, intermediate services and demand.
3. Output: productive entrepreneurship.

**Findings**

- The index explains between four-tenths and six-tenths of the variation in high-growth firms. (p. 827)
- The authors call the tests indications of tendencies, not causal proof. (p. 827)
- Utrecht scores 12.42 on the index and Drenthe 5.99. (p. 826)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Dec 2024 Q1: the ten elements set institutions beside resources and demand. (our reading)
- Practice: ecosystem quality shows in growth-oriented entrepreneurship. (our reading)

**Limits**

- Twelve regions, three points in time, one country. (p. 827)

**Hook**

Ten elements, one system, a score you can compare.

### Paper 8: Autio et al. (2014)

*Entrepreneurial innovation: the importance of context*, Research Policy 43(7).

**In one line**

The editors ask when and where new ventures innovate, and answer with six contexts around the entrepreneur, in time and space.

**Key concepts**

- **Entrepreneurial innovation** (*entreprenøriell innovasjon*): innovation carried out by new ventures, whose rate varies enormously between countries and regions. (p. 1099)
- **Selection effects** (*seleksjonseffekter*): opportunity costs and social legitimacy decide who becomes an entrepreneur. (p. 1100)
- **Strategic choice effects** (*strategiske valgeffekter*): what those who enter then judge desirable and feasible. (p. 1100)
- **Context** (*kontekst*): the industry, organisation, institutions and social setting around the entrepreneur, in time and space. (p. 1100)
- **Innovation system** (*innovasjonssystem*): the literature that explains structures and institutions but studies agency less. (p. 1098)

**Model**

**Six contexts of entrepreneurial innovation** (p. 1100, Fig. 1):

1. Industry and technology.
2. Organisational context.
3. Institutions and policy.
4. Social context.
5. Temporal context.
6. Spatial context.

**Findings**

- Fewer than 30 percent of new ventures report products that are new to their customers, so most are not innovative. (p. 1098)
- Innovation systems research explains structures but sidelines agency and entrepreneurship. (p. 1098)
- The interplay of contexts makes different ecosystems produce different kinds of innovation. (p. 1100)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Dec 2024 Q2: agency works through selection and strategic choice in a context of six settings. (our reading)

**Limits**

- The paper introduces a special issue, so the framework opens a research agenda for others. (p. 1097)

**Hook**

Innovation is not entrepreneurship; the context decides.

### Paper 14: Bækkelund (2021)

*Change agency and reproductive agency in the course of industrial path evolution*, Regional Studies 55(4).

**In one line**

Bækkelund follows three fjord villages and finds reproductive agency doing more than blocking change.

**Key concepts**

- **Industry path** (*næringsbane*): citing Fredin and colleagues, an activity that gathers enough critical mass for self-reinforcing effects. (p. 758)
- **Agency** (*aktørskap*): citing Sotarauta and Suvinen, action or intervention by an actor to produce a particular effect. (p. 758)
- **Change agency** (*endringsaktørskap*): Schumpeterian entrepreneurship, institutional entrepreneurship and place-based leadership. (p. 759)
- **Reproductive agency** (*reproduktivt aktørskap*): the counterparts, replicative entrepreneurship, institutional work and maintenance leadership. (p. 759)
- **Critical juncture** (*kritisk veiskille*): a moment that can move a path into creation, if an opportunity space opens. (p. 765)

**Model**

**A range from change to reproductive agency** (p. 760, Figure 1):

1. Change: Schumpeterian entrepreneurship.
2. Change: institutional entrepreneurship.
3. Change: place-based leadership.
4. Reproductive: replicative entrepreneurship.
5. Reproductive: institutional work.
6. Reproductive: maintenance leadership.

**Findings**

- Reproductive agency is stronger before formation and in path development; change agency is stronger in path creation. (p. 765)
- A shift into path creation often follows a critical juncture with an opportunity space. (p. 765)
- Public actors can enact Schumpeterian entrepreneurship, as in Flåm. (p. 765)

**Norway link**

- **Flåm, Odda and Sogndal**: three fjord villages with the same institutions and landscape but different tourism paths. (p. 761)
- **Flåm**: more change agency than Odda and Sogndal. (p. 765)

**Exam use**

- Dec 2024 Q2: both kinds of agency decide whether a path turns or stays. (our reading)

**Limits**

- The cases come only from rural tourism, an industry with peculiarities of its own. (p. 766)

**Hook**

Two kinds of agency, one path's life.

### Paper 15: Garud & Karnøe (2003)

*Bricolage versus breakthrough: distributed and embedded agency in technology entrepreneurship*, Research Policy 32.

**In one line**

Two wind-power paths show how distributed, embedded agency let modest Danish actors beat far better-funded American ones.

**Key concepts**

- **Distributed agency** (*distribuert aktørskap*): the inputs that shape a technology come from actors in different domains, each with its own frame. (p. 280)
- **Embedded agency** (*innebygd aktørskap*): the accumulating path begins to enable and constrain the actors who built it. (p. 281)
- **Bricolage** (*bricolage*): citing Levi-Strauss, resourcefulness and improvisation with modest, steady gains. (p. 278)
- **Breakthrough** (*gjennombrudd*): an attempt at dramatic outcomes. (p. 279)
- **Technology entrepreneurship** (*teknologisk entreprenørskap*): the process by which a technology and the actors around it shape each other. (p. 277)

**Model**

**The artifact and its embedding processes** (p. 279, Figure 2):

1. Regulation: the rules around the technology.
2. Evaluation: how performance is judged and compared.
3. Design and production: how the artifact is made.
4. Use: how users take it up.

**Findings**

- Danish actors with modest resources became world leaders in wind power, against better-funded American actors. (p. 277)
- Agency is distributed across actors with different interpretive frames. (p. 280)
- The path, once built, enables and constrains the actors who built it. (p. 281)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Dec 2024 Q2: agency is distributed, so a shift needs many actors, not one hero. (our reading)

**Limits**

- The authors say no additional data can remove the interpretive asymmetry between actors. (p. 280)

**Hook**

Modest means, steady gains, and world leadership.

### Paper 16: Miörner & Trippl (2017)

*Paving the way for new regional industrial paths: actors and modes of change in Scania's games industry*, European Planning Studies 25(3).

**In one line**

A constraining region can become enabling when a few actors layer, adapt and reapply what already exists.

**Key concepts**

- **Regional environment** (*regionalt miljø*): pre-existing industry plus the organisational and institutional support built over time. (p. 481)
- **Layering** (*lagdeling*): adding a new element to the support structure, such as a policy or an organisation. (p. 486)
- **Adaptation** (*tilpasning*): changing or reorienting existing institutions and organisations. (p. 486)
- **Novel application** (*ny anvendelse*): using existing structures in new, unforeseen ways. (p. 487)
- **Interpretative, network and institutional power** (*fortolkende, nettverks- og institusjonell makt*): citing Sotarauta and Mustikkamäki, the power an actor holds. (p. 486)

**Model**

**Three modes of change** (p. 486):

1. Layering: adding something new to the structure.
2. Adaptation: changing what exists.
3. Novel application: reusing what exists in a new way.

**Findings**

- A few key individuals combined modes across scales to make the environment enabling. (p. 481)
- The new elements were the Nordic Game Conference, the Game Assembly and Game City. (p. 490)
- Power decides the mode: strong actors layer, weaker ones adapt or reapply. (p. 494)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Jun 2025 Q2: what held the games industry back, and how a few actors changed the support structure. (our reading)

**Limits**

- The findings rest on one case, and the authors do not generalise them. (p. 495)

**Hook**

A closed door, opened by a few hands.

### Paper 9: Hekkert et al. (2007)

*Functions of innovation systems: a new approach for analysing technological change*, Technological Forecasting and Social Change 74(4).

**In one line**

The paper replaces structures with seven activities, and shows how a few can set a system moving.

**Key concepts**

- **Technological innovation system** (*teknologisk innovasjonssystem*): citing Carlsson and Stankiewicz, a network of agents that create, spread and use a technology under shared institutions. (p. 416)
- **Function** (*funksjon*): the activities that produce technological change, studied instead of comparing structures. (p. 415)
- **Guidance of the search** (*retning for søket*): the function that sets expectations and goals, and so steers where a technology goes. (p. 423)
- **Motors of change** (*endringsmotorer*): the few functions that start a virtuous cycle and pull the others along. (p. 426)
- **Vicious cycle** (*ond sirkel*): negative fulfilment in one function that slows the others and can stop progress. (p. 427)

**Model**

**Three motors of change** (p. 426, Figure 2):

1. Motor A: lobbying for market formation sets off entrepreneurial activity and knowledge.
2. Motor B: lobbying for research resources raises expectations about the technology.
3. Motor C: guidance of the search sets goals, then resources and knowledge follow.

**Findings**

- Fulfilling one function affects the others, so the model is non-linear and full of interactions. (p. 426)
- Developments often start from a few functions that pull the rest. (p. 426)
- First Dutch studies found the functions fitted the data well. (p. 428)

**Norway link**

- No Norwegian case in this paper.

**Exam use**

- Jun 2025 Q3: use the seven functions to show where guidance can act. (our reading)
- Dec 2024 Q3: read promotion and inhibition through strong and weak functions. (our reading)

**Limits**

- The authors keep the word function as a heuristic, not as functionalism. (p. 429)

**Hook**

Seven activities, three motors, one moving system.

### Paper 10: Steen et al. (2024)

*CCS technological innovation system dynamics in Norway*, International Journal of Greenhouse Gas Control 136.

**In one line**

The Norwegian CCS system has grown fast, but its market, resources and material base stay weak.

**Key concepts**

- **Technological innovation system** (*teknologisk innovasjonssystem*): actors, networks and institutions that build and spread a technology. (p. 2)
- **Function** (*funksjon*): a process that builds and spreads the technology, eight of them here. (p. 2)
- **Materialization** (*materialisering*): the function for large physical infrastructure, added because CCS needs it. (p. 2)
- **Legitimation** (*legitimering*): the function for acceptance and legitimacy. (p. 3)
- **Market formation** (*markedsforming*): the function that builds a market, rated weak here. (p. 3)

**Model**

**Assessing a technological innovation system** (p. 3):

1. Map the actors, networks and institutions.
2. Rate every function as strong, intermediate or weak.
3. Aim policy at the weak functions.

**Findings**

- Knowledge is strong, but diffusion is limited because projects are customised and firms protect their ideas. (p. 9)
- Market formation, resource mobilization and materialization are weak. (p. 9)
- Legitimacy rises as CCS moves from fossil energy into waste and process industry. (p. 9)

**Norway link**

- **Longship and Northern Lights**: the main exceptions to weak materialization, as large projects being built. (p. 10)
- **Mongstad**: an earlier plan for a full-scale plant at the refinery and power station. (p. 4)

**Exam use**

- Jun 2025 Q3: find the weak functions and aim policy at them. (our reading)

**Limits**

- Other interviewees might have produced slightly different results. (p. 11)
- CCS innovation is international, so a wider geographical lens would help. (p. 11)

**Hook**

Eight functions, one weak chain, a policy target.

### Paper 17: Njøs et al. (2020)

*Expanding analyses of path creation: interconnections between territory and technology*, Economic Geography 96(3).

**In one line**

Two green technologies in the same region sit in different phases, because territory and technology connect differently.

**Key concepts**

- **Path creation** (*baneskaping*): citing MacKinnon and colleagues, the emergence of new development trajectories in a region. (p. 268)
- **Territorial capabilities** (*territorielle kapasiteter*): knowledge sharing, support structures and industrial policy. (p. 266)
- **Guidance of the search, market formation and legitimation** (*retning for søket, markedsforming og legitimering*): the three functions the paper adds to evolutionary economic geography. (p. 271)
- **Preformation** (*preformasjon*): the early phase before a path forms, where the CCS case stays. (p. 281)

**Model**

**Three phases of a path**, citing Martin (p. 268):

1. Preformation: before a path forms.
2. Path creation.
3. Path development.

**Findings**

- Batteries reached green path creation through regional competence, a cluster and public procurement, reaching some seventy ferry lines. (pp. 281–282)
- CCS stays in preformation: few dedicated firms, no market and top-down policy. (p. 281)
- Territory and technology have to be analysed together. (p. 266)

**Norway link**

- **Maritime batteries**: NCE Maritime CleanTech set a direction and public procurement created the market. (pp. 281–282)
- **Technology Centre Mongstad**: the region hosts it, yet CCS stays in preformation. (p. 281)

**Exam use**

- Jun 2025 Q2: batteries were promoted by procurement, a cluster and an agency; CCS was held back. (our reading)

**Limits**

- The authors leave three questions open, including whether green path creation differs from other industries. (p. 284)

**Hook**

Two green technologies, one region, two phases.

### Paper 11: Bargoni et al. (2024)

*Crowdfunding and innovation: a bibliometric review and future research agenda*, European Journal of Innovation Management 27(4).

**In one line**

A bibliometric review maps how crowdfunding research meets innovation research, and sets an agenda for what is still open.

**Key concepts**

- **Crowdfunding** (*folkefinansiering*): collective financing that also works as an instrument to foster innovation. (p. 1110)
- **Incremental innovativeness** (*inkrementell innovativitet*): citing Chan and Parhankangas, a cumulative improvement of existing technology and capabilities. (p. 1115)
- **Radical innovativeness** (*radikal innovativitet*): citing Chan and Parhankangas, change that is more drastic, unique and comprehensive. (p. 1115)
- **Openness** (*åpenhet*): citing Stanko and Henard, seeking ideas from outside, which crowdfunding embodies. (p. 1116)
- **Social capital** (*sosial kapital*): citing Nahapiet and Ghoshal, resources available through a network of relationships. (p. 1117)

**Model**

**The bibliometric review** (p. 1113):

1. Search Scopus and screen with PRISMA.
2. Map co-citations and keyword co-occurrence.
3. Place themes on a centrality and density map.

**Findings**

- Two thousand five hundred and forty-four records were reduced to 150 studies. (p. 1112)
- Co-citation gives three clusters: innovativeness, openness and platform innovation. (p. 1121)
- On the thematic map reward-based crowdfunding is a motor theme, while finance stays emerging. (p. 1120)

**Norway link**
- No Norwegian case in this paper.

**Exam use**
- Dec 2024 Q2: crowdfunding communities as one form of agency in a system. (our reading)

**Limits**
- Scopus was the only source, so work indexed elsewhere may be missing. (p. 1127)
- Articles in languages other than English were excluded. (p. 1127)

**Hook**
One hundred and fifty studies, three clusters, one agenda.

### Paper 12: Demattos Guimarães, Maehle & Bonet (2024)

*The relational forms of cultural-creative crowdfunding: a typology of practices through mapping platforms in Europe and Latin America*, Poetics 105.

**In one line**
A typology of cultural-creative crowdfunding, built from 89 platforms in Europe and Latin America.

**Key concepts**
- **Cultural-creative crowdfunding** (*kultur-kreativ folkefinansiering*): financing culture projects by asking a crowd for contributions through online platforms, a set of diverse practices. (p. 2)
- **Cultural policy paradigms** (*kulturpolitiske paradigmer*): citing Bonet and Négrier, excellence, democratisation, cultural democracy and the creative economy. (p. 4)
- **Tracking and trawling** (*sporing og tråling*): citing Whiting and Prichard, the digital method used to build the database. (p. 6)
- **Typology of practices** (*praksistypologi*): the map from institutional structure to practices, the paper's main output. (p. 12)

**Model**
**Six dimensions of each platform** (p. 8):
1. Country.
2. Geographical scope.
3. Business model.
4. Focus.
5. Institutional structure.
6. Foundation year.

**Findings**
- 67 of the 89 platforms were founded between 2009 and 2015. (p. 9)
- 51 platforms use a hybrid business model, against 38 with a single one. (p. 9)
- In Latin America no platform serves a single sector, while European platforms are more often regional. (p. 12)

**Norway link**
- No Norwegian case in this paper.

**Exam use**
- Dec 2024 Q2: crowdfunding communities as agency that moves resources away from gatekeepers. (our reading)

**Limits**
- The study relies on what platforms display, and that information is inconsistent. (p. 8)
- US platforms such as Kickstarter are left out, and a North American study is invited. (p. 10)

**Hook**
Eighty-nine platforms, six dimensions, one typology of practices.

### Paper 13: Quero & Ventura (2019)

*Value proposition as a framework for value cocreation in crowdfunding ecosystems*, Marketing Theory 19(1).

**In one line**
The authors treat crowdfunding as a service ecosystem and map how actors cocreate its value propositions.

**Key concepts**
- **Crowdfunding ecosystem** (*folkefinansieringsøkosystem*): a service ecosystem in which actors cocreate value to carry one actor's proposal forward. (p. 48)
- **Value proposition** (*verdiforslag*): citing Frow and colleagues, a mechanism that adjusts as actors negotiate how resources are shared. (p. 52)
- **The Eight Co's** (*de åtte co-ene*): the eight ways a value proposition is cocreated, an extension of five earlier categories. (p. 53)
- **Micro, meso and macro levels** (*mikro-, meso- og makronivå*): citing Chandler and Vargo, direct exchange, third-party exchange and networks. (p. 55)

**Model**
**Five steps of value cocreation**, adapting Frow and Payne (p. 51):
1. Identify actors.
2. Determine core values.
3. Facilitate dialogue between the actors.
4. Identify cocreation opportunities.
5. Cocreate value propositions with the actors.

**Findings**
- The two directors added public institutions, a new actor in the ecosystem. (p. 58)
- Across the three levels the authors find eight forms of cocreation in six value proposition metaphors. (p. 60)
- At the macro level, associations and public institutions bring institutionalisation, a possibly conflictual form of cocreation. (p. 58)

**Norway link**
- No Norwegian case in this paper.

**Exam use**
- Dec 2024 Q2: crowdfunding communities as a service ecosystem. (our reading)

**Limits**
- The sampling could broaden the kinds of cultural project and the countries studied. (p. 59)
- Checking the model outside the cultural setting would be interesting. (p. 59)

**Hook**
Eight actors, three levels, one proposal carried forward.

# Part 4: Glossary and sources

## Glossary: English and Norwegian terms

### Terms

| English | Norsk (bokmål) | Where the Norwegian term comes from |
|---|---|---|
| Innovation system | innovasjonssystem | [Class 1, slide 14] |
| National innovation system | nasjonalt innovasjonssystem | [Class 1, slide 2] |
| Regional innovation system | regionalt innovasjonssystem | [Class 2, slide 13] |
| Technological innovation system | teknologisk innovasjonssystem | [Class 6, slide 12] |
| Entrepreneurial ecosystem | entreprenørielt økosystem | [Class 4, slide 1] |
| Crowdfunding ecosystem | folkefinansieringsøkosystem | standard term |
| Institutions | institusjoner | [Class 1, slide 15] |
| Institutional work | institusjonelt arbeid | [Class 3, slide 1] |
| Institutional logics | institusjonelle logikker | standard term |
| Agency | aktørskap | [Class 3, slide 10] |
| Change agency | endringsaktørskap | [Class 3, slide 10] |
| Reproductive agency | reproduktivt aktørskap | [Class 3, slide 10] |
| Path dependency | stiavhengighet | [Class 1, slide 32] |
| Path creation | baneskaping | standard term |
| Path extension | baneforlengelse | standard term |
| Related variety | relatert variasjon | standard term |
| Bricolage | bricolage | standard term |
| Trinity of innovation | treenighet | standard term |
| Function | funksjon | [Class 6, slide 15] |
| Guidance of the search | retning for søket | [Class 6, slide 22] |
| Market formation | markedsforming | standard term |
| Materialization | materialisering | [Class 6, slide 24] |
| Legitimation | legitimering | [Class 6, slide 23] |
| Munificence | rikelighet | standard term |
| Functionality | funksjonalitet | standard term |
| Selection effects | seleksjonseffekter | standard term |
| Strategic choice effects | strategiske valgeffekter | standard term |
| Green industrial restructuring | grønn industriell omstilling | standard term |
| Paradigm | paradigme | standard term |
| Dedicated innovation system | dedikert innovasjonssystem | standard term |
| Transformative innovation | transformativ innovasjon | standard term |
| Transformational failures | transformasjonssvikt | standard term |
| Crowdfunding | folkefinansiering | [Bonus, slide 7] |
| Incremental innovativeness | inkrementell innovativitet | standard term |
| Radical innovativeness | radikal innovativitet | standard term |
| Value proposition | verdiforslag | standard term |

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### Readings


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- Quero & Ventura (2019). Value proposition as a framework for value cocreation in crowdfunding ecosystems. Marketing Theory 19(1), 47–63.


### Classes

- Class 1: Innovation as systemic and sustainable transformation. Arnt Fløysand. Slides: Forlesning 1 Fløysand.pdf (41 slides, H2025).
- Class 2: Regional innovation systems and sustainable transformation. Arnt Fløysand. Slides: Forlesning 2 Fløysand.pdf (44 slides, H2025).
- Class 3: Institutional work in the regional innovation systems of Hordaland and Rogaland. Arnt Fløysand. Slides: Forelesning 5 Fløysand Institusjonelt arbeid.pdf (34 slides, H2025).
- Class 4: Entrepreneurial ecosystems and sustainable transformation. Nora Geirsdotter Bækkelund. Slides: Forelesning3 Entreprenørielle økosystem H2025.pdf (24 slides, H2025).
- Class 5: Agency and changing innovation systems. Nora Geirsdotter Bækkelund. Slides: Forelesning 6 System og aktørskap H2025.pdf (22 slides, H2025).
- Class 6: Technological innovation systems. Rune Njøs (guest lecture). Slides: Forelesning 4 Njøs.pdf (43 slides, H2025).
- Class 9: Summary and exam preparation. Arnt Fløysand. Slides: Forelesning 8 Fløysand Oppsummering.pdf (43 slides, H2025).
- Bonus: The art of crowdfunding. Alice Demattos Guimarães. Slides: INN514 Lecture CF.pdf (50 slides, H2025).

### Exam papers

- INN514 exam paper, 4 December 2024.
- INN514 exam paper, 13 June 2025.

# Part 5: Answer plans

### Answer plans for the six past questions

Drafting aid (ours): learn the moves, then write your own words. Each plan is voiced as a worked answer on the companion site (Revision 11 to 16).

**December 2024, Question 1**

**Question:** What do you understand by innovation system? What do you understand by sustainable transformation? What do you understand by "agency"? What do you understand by "institutions"? **Thesis:** Define each term from the course, name its source, and show one Norwegian case. **Moves:** 1. Define the broad system (Fagerberg et al. 2009, p. 431) 2. Use the UNEP change line (Fløysand et al. 2022, p. 17) 3. Separate two kinds of agency (Bækkelund 2021, p. 758) 4. Call institutions rules of the game (Fagerberg et al. 2009, p. 431) 5. Join them in the two regions (Fløysand et al. 2022, p. 15) **Example:** In Hordaland collective work legitimised a green shift, and in Rogaland individual work did not (Fløysand et al. 2022, p. 15). **Trap:** Do not use a present-needs sentence as the definition of sustainable transformation.

**December 2024, Question 2**

**Question:** How can "agency" in innovation systems promote or inhibit sustainable transformation? Use examples. **Thesis:** Agency can open a greener path or keep an old path in place. **Moves:** 1. Read both sides of the command [Class 9, slide 39] 2. Define both kinds of agency (Bækkelund 2021, p. 758) 3. Show how change work promotes (Bækkelund 2021, p. 758) 4. Show how reproductive work inhibits (Fløysand et al. 2022, p. 22) 5. Use the two regions, then close (Fløysand et al. 2022, p. 15) **Example:** Collective work in Hordaland legitimised a green shift, while individual leadership in Rogaland did not (Fløysand et al. 2022, p. 15). **Trap:** Do not call either kind of agency good or bad.

**December 2024, Question 3**

**Question:** How can technologies in innovation systems promote or inhibit sustainable transformation? Use examples. **Thesis:** A technology helps when direction and a market are built, and hinders when those functions stay weak. **Moves:** 1. Read both sides of the command [Class 9, slide 39] 2. Define the system and functions (Hekkert et al. 2007, p. 421) 3. Show a direction and a market (Njøs et al. 2020, p. 281) 4. Show weak functions that inhibit (Steen et al. 2024, p. 9) 5. Use batteries and capture, then close (Steen et al. 2024, p. 4) **Example:** A cluster set a battery direction, and public procurement created the market (Njøs et al. 2020, p. 281). **Trap:** Do not treat knowledge and legitimation as strengths of capture.

**June 2025, Question 1**

**Question:** What do you understand by a regional innovation system? What do you understand by institutional work? What do you understand by sustainability? **Thesis:** Define each term from the course, name its source, and show the two regions. **Moves:** 1. Define the regional system (Fløysand et al. 2022, p. 16) 2. Name the four trajectories (Fløysand & Jakobsen 2016, p. 328) 3. Define institutional work (Fløysand et al. 2022, p. 15) 4. Use the three UN lines [Class 9, slide 29] 5. Show the two directions of work (Fløysand et al. 2022, p. 22) **Example:** Collective work in diversified Hordaland legitimised a green shift, unlike Rogaland (Fløysand et al. 2022, p. 15). **Trap:** Do not use the present-needs sentence as the definition.

**June 2025, Question 2**

**Question:** In what way can a regional innovation system promote or inhibit industrial restructuring? Use examples. **Thesis:** A regional system promotes restructuring through variety and collective work, and inhibits it when specialised. **Moves:** 1. Read both sides of the command [Class 9, slide 39] 2. Name the four trajectories (Fløysand & Jakobsen 2016, p. 328) 3. Show how variety can promote (Fløysand & Jakobsen 2016, p. 330) 4. Show how lock-in can inhibit (Fløysand & Jakobsen 2016, p. 328) 5. Use the two regions, then close (Fløysand et al. 2022, p. 15) **Example:** Maritime batteries are embedded, and a cluster plus procurement built the market (Njøs et al. 2020, p. 281). **Trap:** Keep exhaustion in the four names. Upgrading means renewal.

**June 2025, Question 3**

**Question:** How can technological innovation systems be guided to promote sustainable transformation? Use examples. **Thesis:** Guide the system by rating functions, aiming policy at the weak ones, and choosing the direction. **Moves:** 1. Read the demand for levers [Class 9, slide 39] 2. Define the seven functions (Hekkert et al. 2007, p. 421) 3. Aim policy at the weak ones (Steen et al. 2024, p. 9) 4. Choose the direction on purpose (Schlaile et al. 2017, p. 4) 5. Use capture and batteries, then close (Steen et al. 2024, p. 4) **Example:** Longship adds two capture sites, ship transport and offshore storage (Steen et al. 2024, p. 4). **Trap:** Do not call knowledge and legitimation strengths of the capture system.